Listing 17.1 Auto Scaling group and launch template for a web app

```
# [...]
LaunchTemplate:
  Type: 'AWS::EC2::LaunchTemplate'
  Properties:
    LaunchTemplateData:
      IamInstanceProfile:
        Name: !Ref InstanceProfile
      ImageId: 'ami-028f2b5ee08012131'          ①
      InstanceType: 't2.micro'                  ②
      NetworkInterfaces:
      - AssociatePublicIpAddress: true          ③
        DeviceIndex: 0
        Groups:                                 ④
        - !Ref WebServerSecurityGroup
      UserData:                                 ⑤
        'Fn::Base64': !Sub |
          #!/bin/bash -x
          yum -y install httpd
AutoScalingGroup:
  Type: 'AWS::AutoScaling::AutoScalingGroup'
  Properties:
    TargetGroupARNs:                            ⑥
    - !Ref LoadBalancerTargetGroup
    LaunchTemplate:                             ⑦
      LaunchTemplateId: !Ref LaunchTemplate
      Version: !GetAtt 'LaunchTemplate.LatestVersionNumber'
    MinSize: 2                                  ⑧
    MaxSize: 4                                  ⑨
    HealthCheckGracePeriod: 300                 ⑩
    HealthCheckType: ELB
    VPCZoneIdentifier:
    - !Ref SubnetA
    - !Ref SubnetB
# [...]
```

① Image (AMI) from which to launch new virtual machines

② Instance type for new EC2 instances

③ Associates a public IP address with new virtual machines

④ Attaches these security groups when launching new virtual machines

⑤ The script executed during the bootstrap of virtual machines

⑥ Registers new virtual machines on the target group of the load balancer

⑦ References the launch template

⑧ Minimum number of EC2 instances

⑨ Maximum number of EC2 instances

⑩ Waits 300 seconds before terminating a new virtual machine because of an unsuccessful health check

Uses the health check from the ELB to check the health of the EC2 instances

Starts the virtual machines in these two subnets of the VPC